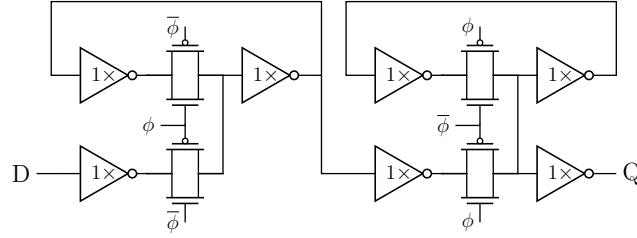


EE5311 Digital IC Design: Assignment 6
Setup time and Clock-Q delay of D flip-flop

- Using a $1\times$ inverter ($(\frac{W}{L})_n = \frac{0.42\mu\text{m}}{0.15\mu\text{m}}, (\frac{W}{L})_p = \frac{0.84\mu\text{m}}{0.15\mu\text{m}}$), draw the below schematic of a positive edge-triggered D flip-flop in **xscem**. Size of n and pMOS in the transmission gate: $\frac{W}{L} = \frac{0.42\mu\text{m}}{0.15\mu\text{m}}$.



Measure the setup time (t_{setup}) and clock-Q delay (t_{CQ}) for the above flip-flop **driving a $1\times$ inverter** for the following transitions at Q:

- rising
- falling

Assume that the clock ϕ and its complement $\bar{\phi}$ are available as inputs. Use the following waveform for:
 ϕ : pulse 0 1.8 10n 5p 5p 9.99n 20n and
 $\bar{\phi}$: pulse 1.8 0 10n 5p 5p 9.99n 20n.

Assume 5ps rise and fall times for the D input. Sweep the transition time of D and find the time at which t_{DQ} is minimum. t_{DC} and t_{CQ} for this transition time gives the setup time and clock-Q delay.

Ngspice manual for reference: <https://ngspice.sourceforge.io/docs/ngspice-manual.pdf>